



Real Estate Rental Trends : **Clustering for Insights**

Course Name:

Advanced Topics in Machine Learning

Lecturer Name:

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Problem Statement

- Finding affordable housing in major cities has become extremely challenging. Lack of transparency in pricing factors for the rental market.
- **Goal:** To Uncover new and unexpected links and patterns between the qualities of the apartments, the time dimension, the geographical location, and the rental rates in Tel Aviv-Jaffa.

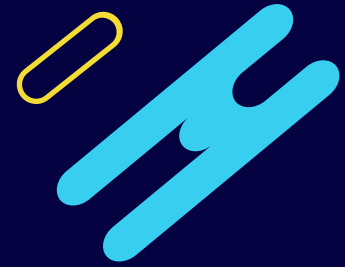


Other ML Analytical Techniques

Methods/techniques related to the problem's domain

- **Natural Language Processing (NLP):** Extracts valuable information from textual data in listings, such as property descriptions, to enhance feature selection and improve model accuracy.
- **Anomaly Detection:** Identifies unusual rental listings or price points, flagging potential outliers for further investigation or revealing unique market opportunities.

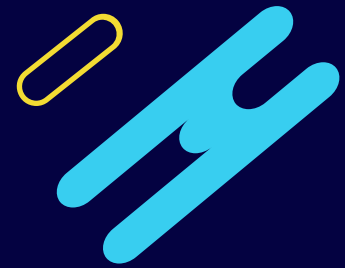




Data Set and Feature Selection

- Developed an automated programming interface using the 'Requests' and 'BeautifulSoup' libraries to scrape ~275,000 rental listings in Tel Aviv-Jaffa from the last decade.
- Performed extensive data cleaning, handling missing values, removing irrelevant features, encoding text features, creating new composite features based on correlations, and more
- Created composite features based on correlation(e.g. 'Price_per_sqm', 'Size_per_room', 'Season', 'Number of rooms')

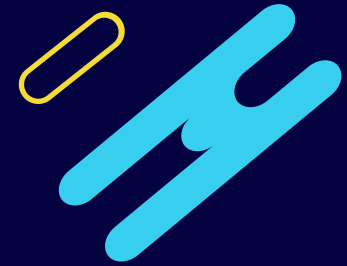




Data Set and Feature Selection

Examples of our data set:

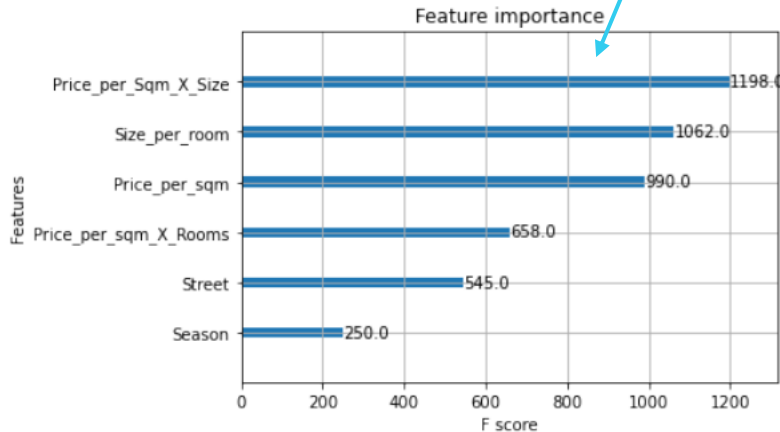
Street	Address	Number of Rooms	Size	Date Post	Price	Floor	Property Tax	House Committee	מרפסת	חניה	מעלית	מזגן	מחסן	מושפעת	מרפסת שמש
מעוז אביב א	בני אפרים 207	3	96	2023-02-01 00:00:00	9000	מתוך 1 3	400	100	0	0	0	1	0	0	0
הצפון הישן החלק המרכזי	ארבע ארצות 9	2	65	2023-02-01 00:00:00	8250	מתוך 3 5	100	200	1	0	1	1	0	1	1
יפו ד	שדרות ירושלים 211	3	60	2023-02-01 00:00:00	4300	מתוך 2 3	160	50	0	1	0	1	0	0	0
התקווה	הרטוב 11	3	65	2023-02-01 00:00:00	4700	מתוך 4 4	175	80	0	1	0	1	0	1	0
נוה שרת	אח"י דקר 25	4	95	2023-02-01 00:00:00	6500	מתוך 1 4	250	225	0	0	1	1	0	1	0
נר אביב	צלילי חנינא 34	3	80	2023-02-01 00:00:00	5900	מתוך 4 4	200	80	1	0	0	1	0	1	1
הצפון הישן החלק הצפוני	עובדיה מברטנורה 16	3.5	75	2023-02-01 00:00:00	7300	מתוך 1 4	250	80	0	0	0	1	0	0	0
הצפון החדש החלק הצפוני	שדרות סמאטס 10	3	92	2023-02-01 00:00:00	9000	מתוך 3 3	420	190	0	0	1	1	0	0	0
הצפון הישן החלק המרכזי	דיזנגוף 214	3	80	2023-02-01 00:00:00	11500	מתוך 6 7	400	150	0	0	1	1	0	1	0
הצפון החדש סביבת כנר המדינה	תש"ח	3.5	80	2023-03-01 00:00:00	10000	מתוך 1 5	435	550	1	0	1	1	0	0	1
בית שטראוס	מוהליבר	3.5	95	2023-03-01 00:00:00	9000	מתוך 2 3	225	150	1	0	0	1	0	0	1
גני צהלה	ענתות 39	4	105	2023-03-01 00:00:00	11500	מתוך 1 4	500	200	1	1	1	1	0	1	1
הצפון החדש סביבת כנר המדינה	ז'בוטינסקי 138	3	95	2023-03-01 00:00:00	8500	מתוך 1 4	450	190	0	0	0	1	0	0	0
רמת אביב	ד"ר חיים ברודי 9	2.5	90	2023-03-01 00:00:00	5400	מתוך 2 2	180	60	1	0	0	1	0	0	1
אזורי חן	שאלו אביגור 7	5	170	2023-03-01 00:00:00	12000	מתוך 3 8	1000	1020	1	1	1	1	0	1	1
יפו ד	הרב רובינשטיין 55	3	70	2023-03-01 00:00:00	4430	מתוך 1 7	205	170	0	0	1	1	0	1	0
נוה שרת	רמה 22	2.5	47	2023-03-01 00:00:00	4500	מתוך 2 4	175	60	0	0	0	1	0	1	0
לב תל אביב החלק הצפוני	אלנבי 1	2	50	2023-03-01 00:00:00	12000	מתוך 18 26	200	1700	0	1	1	1	0	1	0
לב תל אביב החלק הדרומי	שדרות רוטשילד	3	270	2023-03-01 00:00:00	30000	מתוך 3 3	625	1000	1	0	1	1	0	1	1
הצפון הישן החלק הדרום מזרחי	שדרות בן גוריון	1	22	2023-03-01 00:00:00	3700	מתוך 1 4	85	60	1	0	0	1	0	0	1
הצפון הישן החלק המרכזי	אמסטרדם 11	2	65	2023-05-01 00:00:00	7500	קרקע מתוך 4	240	129	1	0	1	1	0	1	1
הצפון הישן החלק הדרום מזרחי	אבן גבירול 39	3	70	2023-07-01 00:00:00	7500	מתוך 3 4	475	150	0	0	1	1	0	0	0
לב תל אביב החלק הדרומי	לבונטין 24	2	50	2023-02-02 00:00:00	9000	מתוך 3 5	210	350	0	0	0	0	0	0	0
כרם התימנים	גאולה 14	2	62	2023-02-02 00:00:00	12400	מתוך 7 9	220	884	1	1	1	1	0	0	1
נוה אביבים	איינשטיין 21	3	75	2023-02-02 00:00:00	7000	מתוך 1 2	275	150	1	1	0	1	0	1	1
הצפון הישן החלק הצפוני	זכריה 20	3	70	2023-02-02 00:00:00	7000	מתוך 1 3	120	100	1	0	0	1	0	0	1
הצפון החדש החלק הצפוני	לואי מרשל 1	2	70	2023-02-02 00:00:00	6400	קרקע מתוך 4	275	100	1	0	0	1	0	0	1
רמת אביב	ברודקיי 35	2	50	2023-02-02 00:00:00	5800	מתוך 2 3	154	90	0	0	0	1	0	1	0
בית שטראוס	צדוק הכהן 3	2	40	2023-02-02 00:00:00	5300	מתוך 1 1	137	150	1	0	0	1	0	0	1



Data Set and Feature Selection

	Feature	Importance
2	Price_per_Sqm_X_Size	0.969336
1	Size_per_room	0.016459
3	Price_per_sqm_X_Rooms	0.008304
0	Price_per_sqm	0.003707
4	Season	0.001198
5	Street	0.000995

Using F1 Score, and XGBoost for importance features



Parameter Selection:

- K-Means: K=4 using Elbow Method and Silhouette Score
- DBSCAN: epsilon=0.5, minPts=15 using KNN
- Hierarchical: Height ~140 based on dendrogram

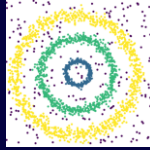
Clustering Algorithms & Techniques



K-Means

Parameter
Selection:

K=4 using the
Elbow Method and
Silhouette Score



DBSCAN

Epsilon=0.5,
MinPts=15 using KNN



Hierarchical Clustering

Height ~140 based on
Dendrogram

Clustering Algorithms & Techniques

• Why Choosing K-Means:

- Suitable for linear/simple patterns in data features like 'price per sqm', 'size per room' etc.
- Efficient and scalable for large datasets by finding cluster centers
- Simplicity makes it a good first clustering approach

• Why Choosing DBSCAN:

- Identifies clusters of varying densities and arbitrary shapes
- Robust to noise and outliers by classifying them separately
- Useful for uncovering complex non-linear patterns in features

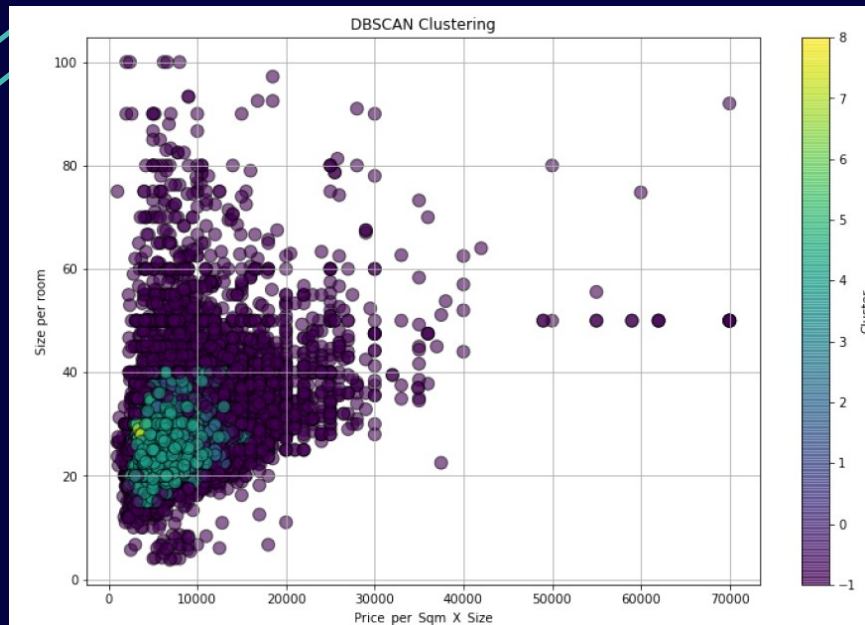
Clustering Algorithms & Techniques

• Why Choosing Hierarchical Clustering:

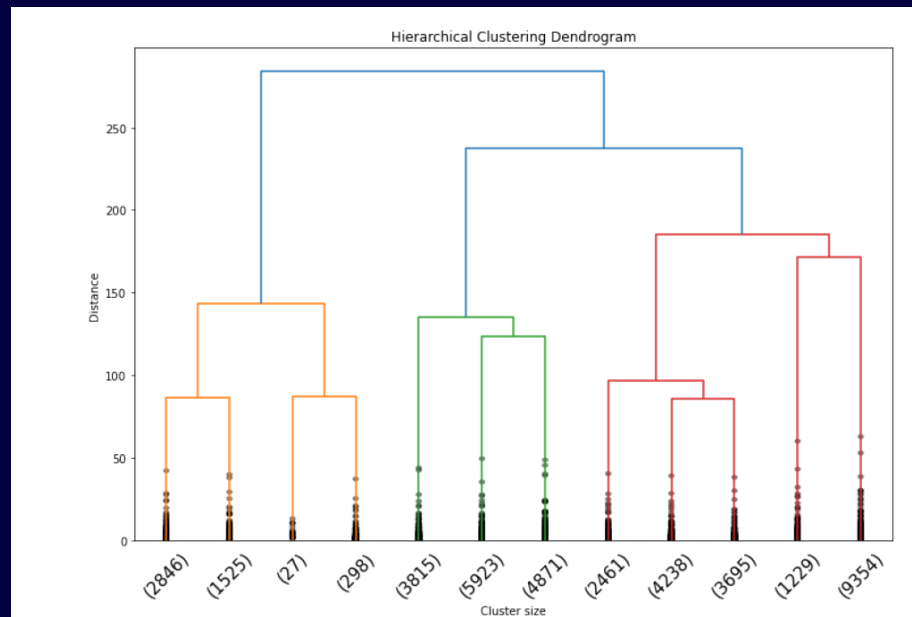
- Provides multi-level views into data hierarchy and similarities
- Dendrograms visualize how data points merge into natural clusters
- Useful for exploring hierarchies within property segments

By combining these algorithms, the goal was to overcome the limitations of any single method and gain comprehensive insights from complementary clustering perspectives tailored to the unique characteristics of the real estate data.

Clustering Algorithms & Techniques

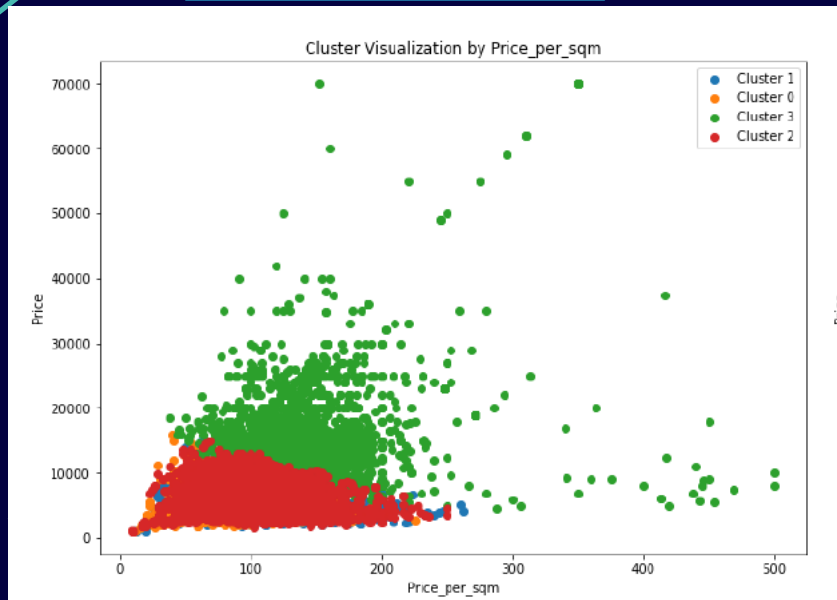


2	10612
4	9964
0	8174
-1	6178
1	4675
5	236
7	217
3	190
8	27
6	9

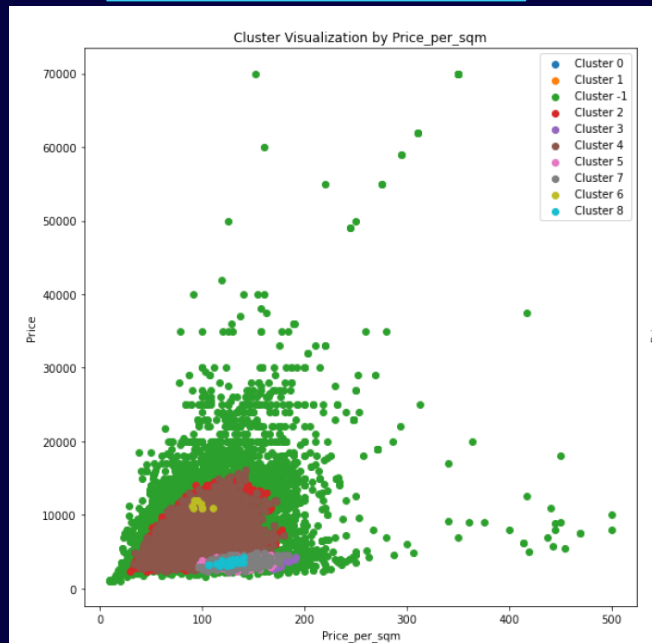


Clustering Algorithms & Techniques

K-Means

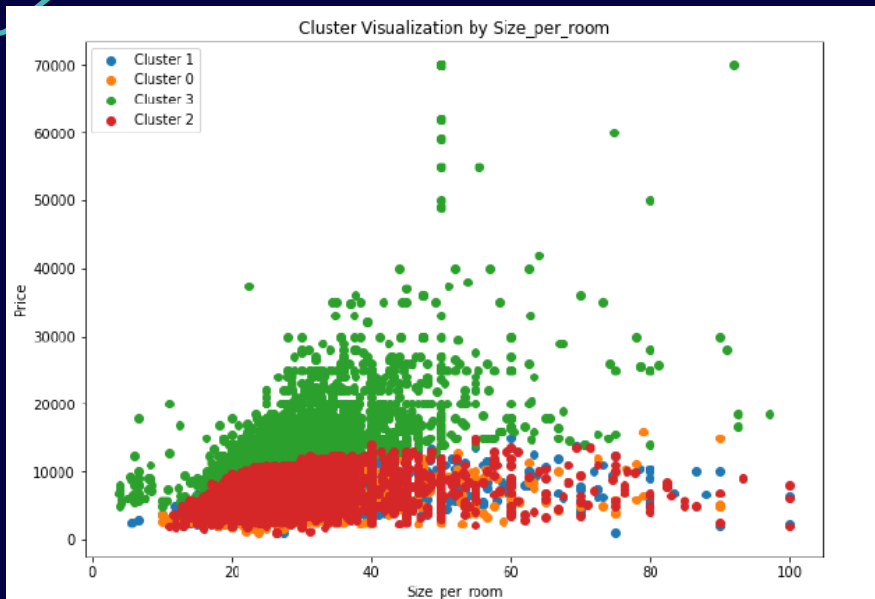


DBSCAN

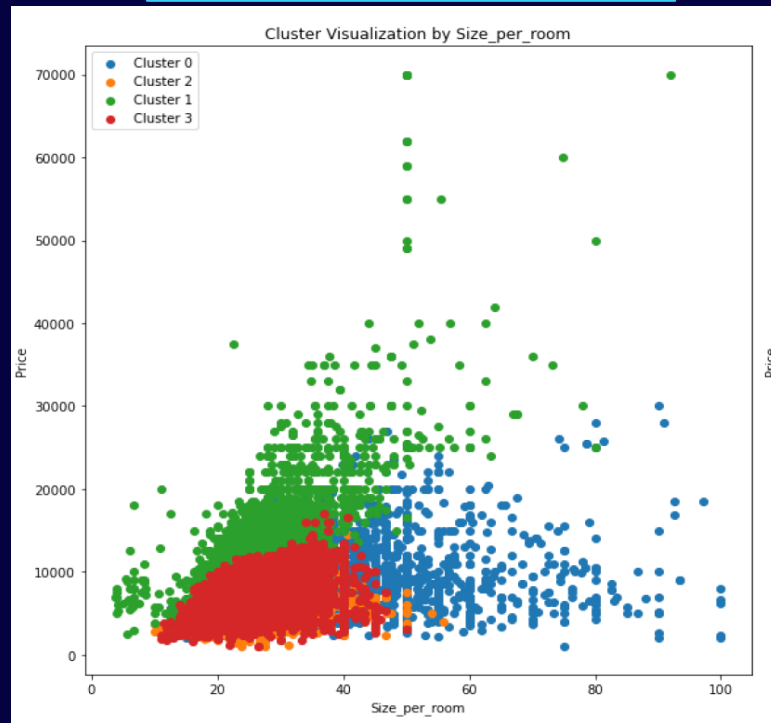


Clustering Algorithms & Techniques

K-Means

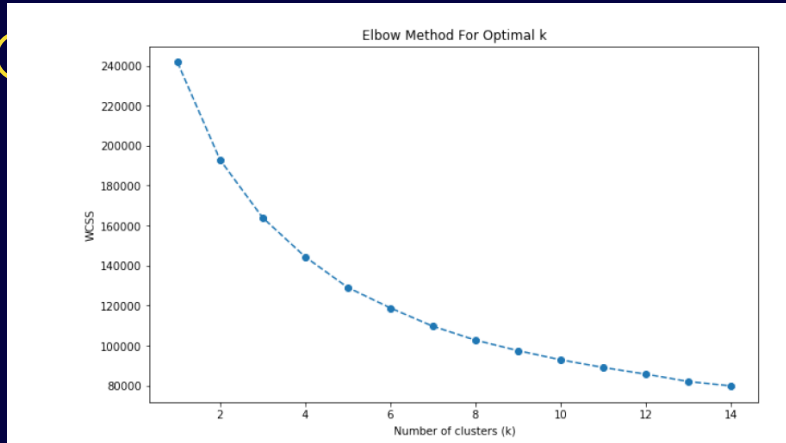


Hierarchical Clustering



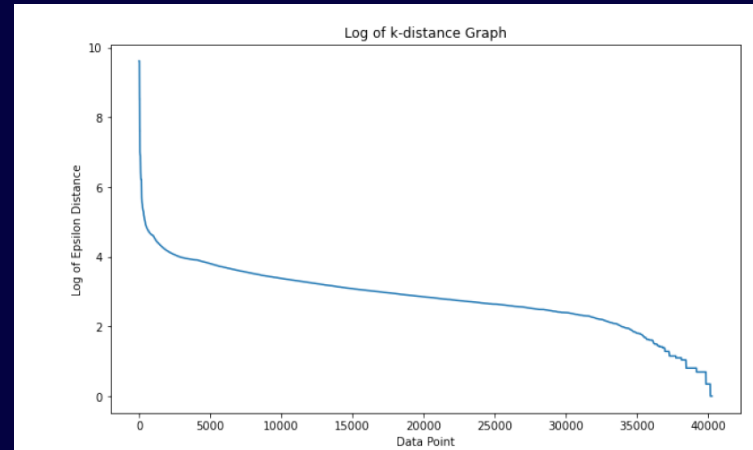
Clustering Algorithms & Techniques

K-Means - Elbow Method



Visual analysis confirmed that the K's that might fit are 4/5/6. After checking each of the K's with the 'Silhouette Score' evaluation tool, we determined that the most appropriate K is K=4, based on the highest score.

DBSCAN - KNN for Epsilon



First, we used a K-NN evaluation procedure, and we got an average epsilon of 148.41, which was too large for the parameter. Then we choose epsilon = 0.5.

Evaluation Metrics

- Silhouette Score
- Davies-Bouldin Index (DBI)
- F1 Score for feature importance
- **K-Means performed best (highest Silhouette (0.5532), lowest DBI (0.5497))**

	K-Means	DBSCAN	Hierarchical Clustering
Silhouette Score	0.5532	0.4875	0.3375
Davies-Bouldin Index	0.5497	2.7431	0.8557

Conclusions and Contributions

01

Gained a deep understanding of rental market dynamics

02

Clustering unveiled new patterns in pricing factors

03

Insights valuable for real estate stakeholders



Future Work

01

Real-time analysis

02

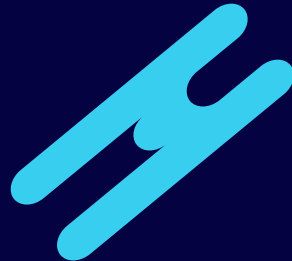
Advanced machine learning for forecasting

03

Spatial analysis with geographic data

04

Studying environmental/social impact factors



Future Work

05

Reference to appraisals

06

Investigating the impact of new technologies

07

Optimization and tuning - fine-tuning algorithms

